

Technical Evaluation Task

AI-Based Resume Analyzer & Job Alignment System

Objective

The objective of this assignment is to evaluate the candidate's ability to:

- Design and build a full-stack web application
- Implement backend APIs
- Apply basic AI integration
- Work with structured and unstructured data
- Design scalable and clean system architecture
- Write clean, maintainable code

Problem Statement

Build a web-based application where:

1. A user can upload a resume (PDF/DOCX).
2. The system extracts key skills and structured information from the resume.
3. The system identifies weak points or missing sections in the resume.
4. The user can paste a job description (JD).
5. The system compares the resume with the JD and generates:
 - a. Skill match percentage
 - b. Missing required skills
 - c. Overall alignment score
 - d. Improvement suggestions

Technical Requirements

Frontend

- React.js with Next.js
- Clean UI with proper state management
- File upload functionality
- Display analysis results clearly

Backend

- NodeJS or Python
- REST APIs
- Resume parsing logic
- User authentication and authorization

Database

- MongoDB or PostgreSQL
- Store user data, resumes, and analysis results

Functional Requirements

The application must:

- Accept resume uploads (PDF/DOCX).
- Extract skills, education, and experience.
- Detect resume weaknesses (e.g., missing sections, lack of measurable achievements, skill gaps).
- Accept job description input.
- Compute a match score between resume and JD.
- Highlight missing skills from JD.

Expected Deliverables

1. Source code (GitHub repository)
2. README with:
 - a. Setup instructions
 - b. Architecture explanation
 - c. Assumptions made
3. API documentation
4. Short demo video (5–10 minutes)
5. Deployed link